

OPERATIONS & EXPRESSIONS

Q1.

SIMATS Online Programming Lab

C PROGRAMMING

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PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main() {
3     int m, n, temp;
4     scanf("%d %d", &m, &n);
5     printf("Before swapping:\n m=%d, n=%d\n", m, n);
6     temp=m;
7     m=n;
8     n=temp;
9     printf("After swapping:\n m=%d, n=%d", m, n);
10    return 0;
11 }
```

INPUT

2
3

OUTPUT

```
Before swapping:
m=2, n=3
After swapping:
m=3, n=2
```

Q2.

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PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main() {
3     int a, b, s, d, p, q;
4     scanf("%d%d", &a, &b);
5     s=a+b;
6     d=a-b;
7     p=a*b;
8     q=a/b;
9     printf("Sum = %d\nDifference=%d\nProduct=%d\nQuotient=%d", s,d,p,q);
10    return 0;
11 }
```

INPUT

4
2

OUTPUT

```
Sum = 6
Difference=2
Product=8
Quotient=2
```

Q3.

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C

Run

Save

```
1 #include<stdio.h>
2 int main() {
3     float celsius, fahrenheit;
4     scanf("%f", &celsius);
5     fahrenheit=(celsius*1.8)+32;
6     printf("Fahrenheit = %f", fahrenheit);
7     return 0;
8 }
```

INPUT

\$35

OUTPUT

```
Fahrenheit = 95.000000
```

Q4.

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C

Run

Save

```
1 #include<stdio.h>
2 int main() {
3     float l, w, area;
4     scanf("%f %f", &l, &w);
5     area=l*w;
6     printf("Area of rectangle = %f", area);
7     return 0;
8 }
```

INPUT

8
4.5

OUTPUT

```
Area of rectangle = 36.000000
```

Q5.

SIMATS Online Programming Lab

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C

Run

Save

```
1 #include<stdio.h>
2 int main() {
3     int a;
4     scanf("%d", &a);
5     if(a%2==0)
6         printf("%d is even", a);
7     else
8         printf("%d is odd", a);
9     return 0;
10 }
```

INPUT

5

OUTPUT

```
5 is odd
```

Q6.

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PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main() {
3     int year;
4     scanf("%d", &year);
5     if ((year%4 == 0 && year%100 != 0) || (year%400 == 0))
6         printf("%d is a leap year", year);
7     else
8         printf("%d is not a leap year", year);
9     return 0;
10 }
```

INPUT

2026

OUTPUT

```
2026 is not a leap year
```

Q7.

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PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main() {
3     int a, b, c, max, min;
4     scanf("%d %d %d", &a, &b, &c);
5
6     if(a>b && a>c)
7         max=a;
8     else if(b>c && b>a)
9         max=b;
10    else
11        max=c;
12    printf("Maximum = %d\n", max);
13
14    if(a<b && a<c)
15        min=a;
16    else if(b<c && b<a)
17        min=b;
18    else
19        min=c;
20    printf("Minimum = %d", min);
21
```

INPUT

6
5
9

OUTPUT

```
Maximum = 9
Minimum = 5
```

Q8.

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PROGRAM EDITOR

C ▼

Run Save

```
1 #include<stdio.h>
2 int main() {
3     int a, b, c, sum, avg;
4     scanf("%d %d %d", &a, &b,&c);
5     sum = a+b+c;
6     avg = sum/3;
7     printf("Sum = %d\nAverage = %d", sum, avg);
8     return 0;
9 }
```

INPUT

2
3
4

OUTPUT

Sum = 9
Average = 3